

Derivation of Growth-Factor Bounds for Pivoting Methods 3

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Section 3 of the Pivoting and Scaling Policies for Half-Precision LU Factorization paper rests on a *single* inequality. Each pivoting rule just plugs in a different choice of two quantities, a multiplier bound μ_k and a pivot-row tail bound α_k (both defined in the Setup below). What follows derives every displayed formula in Section 3, in the order they appear in the paper.

Setup

At elimination step k :

$$\begin{aligned} \text{multiplier: } m_{ik} &= a_{ik}^{(k)} / a_{kk}^{(k)}, \quad i > k, \\ \text{Schur update: } a_{ij}^{(k+1)} &= a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)}, \quad i, j > k, \\ \text{active maximum: } L_k &= \max_{i,j \geq k} |a_{ij}^{(k)}|. \end{aligned}$$

The growth factor of strategy $\star$ is $\rho_\star = \max_k L_k / L_1$.

Two per-step quantities drive every bound below: a *multiplier bound* μ_k with $|m_{ik}| \leq \mu_k$ for $i > k$, and a *pivot-row tail bound* α_k with $\max_{j > k} |a_{kj}^{(k)}| \leq \alpha_k L_k$. The induced *local growth factor* is $g_k = 1 + \mu_k \alpha_k$; this is the same g_k that appears in the Stability Template of the main paper, where $L_{k+1} \leq g_k L_k$.

1 The single mechanism (Section 3.1)

Take absolute values of the Schur update and apply the triangle inequality:

$$|a_{ij}^{(k+1)}| \leq |a_{ij}^{(k)}| + |m_{ik}| |a_{kj}^{(k)}|.$$

Suppose at step k we have two bounds:

$$|m_{ik}| \leq \mu_k, \quad \max_{j > k} |a_{kj}^{(k)}| \leq \alpha_k L_k.$$

Then $|a_{ij}^{(k+1)}| \leq L_k + \mu_k \alpha_k L_k$, and taking the maximum over $i, j > k$:

One-step bound.

$$L_{k+1} \leq (1 + \mu_k \alpha_k) L_k = g_k L_k.$$

Product bound. Iterating from $k = 1$ to $n - 1$,

$$\rho_\star \leq \prod_{k=1}^{n-1} g_k = \prod_{k=1}^{n-1} (1 + \mu_k \alpha_k).$$

The two error inequalities at the end of Section 3.1 (the $\eta_A(\hat{x}) \lesssim C_n u \rho$ backward bound and the $\kappa_\infty(A) C_n u \rho$ forward bound) are the standard first-order Wilkinson–Higham estimates; the paper quotes them rather than re-deriving them.

Recipe. For each pivoting rule we just need

- a multiplier bound μ_k , fixed by the pivot-selection test (e.g. $\mu_k = 1/\tau$ on a successful ratio test, $\mu_k \leq 1$ otherwise);
- a pivot-row tail bound α_k , either the trivial $\alpha_k = 1$ (which holds whenever L_k bounds the whole active block) or the geometric $\alpha_k = \sqrt{c_k^{-2} - 1}$ derived for GP in the Geometric Pivoting section.

Everything else in Section 3 is filling in this recipe: the sections below compute μ_k and α_k for each rule.

2 Delayed Pivoting and Eq. (1)

Throughout, $\tau > 1$ denotes the pivot-acceptance threshold; how its value is chosen is derived in Section 3. Let $M_1 \geq M_2 \geq 0$ denote the largest and second-largest magnitudes in the active part of the chosen pivot column.

When the ratio test succeeds ($M_1/M_2 \geq \tau$). Row partial pivoting moves M_1 to the diagonal, so $|a_{kk}^{(k)}| = M_1$ and $|a_{ik}^{(k)}| \leq M_2$ for $i > k$. Hence

$$|m_{ik}| = \frac{|a_{ik}^{(k)}|}{|a_{kk}^{(k)}|} \leq \frac{M_2}{M_1} \leq \frac{1}{\tau}.$$

The pivot-row tail needs no separate test. Since $L_k = \max_{i,j \geq k} |a_{ij}^{(k)}|$ is the maximum over the *entire* active block, every pivot-row entry satisfies $\max_{j > k} |a_{kj}^{(k)}| \leq L_k$, i.e. the trivial bound $\alpha_k = 1$. This holds regardless of the column ratio test, which only controls the multiplier μ_k . So $\mu_k = 1/\tau$, $\alpha_k = 1$, and the engine gives

$$g_k = 1 + \mu_k \alpha_k = 1 + \frac{1}{\tau}.$$

When the ratio test fails (fallback to PP). Only $|m_{ik}| \leq 1$ is guaranteed, hence $g_k = 2$.

Combining. If s of the $n - 1$ steps succeed and $n - 1 - s$ fall back, the product bound is

$$\rho_{\text{DP}} \leq \left(1 + \frac{1}{\tau}\right)^s 2^{n-1-s}. \quad (\text{Eq. 1})$$

The simpler form $(1 + 1/\tau)^{n-1}$ is the special case $s = n - 1$. Writing the mixed bound is the honest version: in practice some steps do fall back.

3 Where does τ come from?

The threshold τ is set so that an observed top-two ratio is large enough for the true ranking to be reliable.

Step A – separation under componentwise rounding. Two true positive magnitudes x, y are observed in floating point as

$$\hat{x} = x(1 + \delta_x), \quad \hat{y} = y(1 + \delta_y), \quad |\delta_x|, |\delta_y| \leq \eta_r.$$

From $\hat{x} \leq x(1 + \eta_r)$ and $\hat{y} \geq y(1 - \eta_r)$,

$$\frac{x}{y} \geq \frac{\hat{x}}{\hat{y}} \frac{1 - \eta_r}{1 + \eta_r}.$$

Hence $\hat{x}/\hat{y} > (1 + \eta_r)/(1 - \eta_r)$ guarantees $x > y$. This is the meaning of τ : *the observed winner must beat the runner-up by more than the worst-case rounding perturbation that could reverse their true order.*

Step B – deterministic η_r is too pessimistic. The standard worst-case scale is $\eta_r = \gamma_r = ru/(1 - ru)$. With FP16 unit roundoff $u = 2^{-11}$, the condition $\gamma_r < 1$ already fails for $r \geq 1024$, which is below the per-entry rounding history at $n = 2048$. The deterministic version cannot set a useful threshold here.

Step C – probabilistic η_r . Following Higham–Mary (2019) and Connolly–Higham–Mary (2021), mean-zero rounding gives the typical scale

$$\eta_r \approx \sqrt{r} u.$$

At $r = 2048$, $\eta_r \approx 0.0221$, and

$$\tau \approx \frac{1 + 0.0221}{1 - 0.0221} \approx 1.045.$$

This is a model-based separation scale, not a worst-case theorem. The empirical sweet spot $1.005 \leq \tau \leq 1.020$ sits in the same regime, which is why $\tau = 1.01$ is the default.

4 Geometric Pivoting and Eq. (2)

For the chosen pivot row at step k , define the axis alignment

$$c_k = \frac{|a_{kk}^{(k)}|}{\|A_{k,k:n}^{(k)}\|_2}.$$

c_k is the cosine of the angle between the active row and the pivot axis; because of the absolute value in the numerator, $c_k \in (0, 1]$, so it is always nonnegative.

Step A – pivot-row tail in terms of c_k . Pythagoras gives

$$\|A_{k,k:n}^{(k)}\|_2^2 = |a_{kk}^{(k)}|^2 + \|A_{k,k+1:n}^{(k)}\|_2^2.$$

Combined with $\|A_{k,k:n}^{(k)}\|_2 = |a_{kk}^{(k)}|/c_k$,

$$\|A_{k,k+1:n}^{(k)}\|_2 = |a_{kk}^{(k)}| \sqrt{c_k^{-2} - 1}.$$

Since $\max |\cdot| \leq \|\cdot\|_2$ and $|a_{kk}^{(k)}| \leq L_k$,

$$\max_{j>k} |a_{kj}^{(k)}| \leq L_k \sqrt{c_k^{-2} - 1}, \quad \text{so} \quad \alpha_k = \sqrt{c_k^{-2} - 1}.$$

This exposes a clear limit of the geometric bound: when the pivot row is nearly orthogonal to the pivot axis, $c_k \rightarrow 0$, hence $c_k^{-2} \rightarrow \infty$ and $\alpha_k \rightarrow \infty$. The bound is therefore informative only when c_k is bounded away from 0 (a well-aligned pivot row); otherwise one falls back to the trivial $\alpha_k = 1$ from $\max_{j>k} |a_{kj}^{(k)}| \leq L_k$, recovering a PP-like step.

Step B – multiplier bound. GP either (i) picks the clear column maximum, giving $\mu_k \leq 1$, or (ii) in a near tie ($M_1/M_2 < \tau$) picks one of the two largest candidates, giving $\mu_k \leq M_1/M_2 < \tau$. So $\mu_k < \tau$ in either case.

Step C – engine.

$$L_{k+1} \leq \left(1 + \mu_k \sqrt{c_k^{-2} - 1}\right) L_k.$$

Iterating with $c_{\min} = \min_k c_k$:

$$\rho_{\text{GP}} \leq \prod_{k=1}^{n-1} \left(1 + \mu_k \sqrt{c_k^{-2} - 1}\right) \leq \left(1 + \tau \sqrt{c_{\min}^{-2} - 1}\right)^{n-1}. \quad (\text{Eq. 2})$$

When does this beat PP? The per-step bound $1 + \mu_k \sqrt{c_k^{-2} - 1} < 2$ rearranges to

$$c_k > \frac{\mu_k}{\sqrt{1 + \mu_k^2}}.$$

This is why the paper labels Eq. (2) *diagnostic*: it is sharper than PP exactly when the chosen pivot row is sufficiently aligned with the pivot axis, and reduces to a PP-like bound otherwise.

5 Scattered Pivoting

Once ScaP swaps a sampled column into position k , ordinary row partial pivoting takes over inside that column, so $|m_{ik}| \leq 1$. The pivot-row tail again satisfies $\alpha_k = 1$. So $\mu_k = \alpha_k = 1$ and $g_k = 2$:

$$\rho_{\text{ScaP}} \leq 2^{n-1}.$$

This is the same deterministic bound as PP. The advantage of ScaP is empirical sampling diversity, not a stronger worst-case guarantee, and the paper states this explicitly.

6 Dynamic Scaling

Power-of-two scaling is exact. A normal binary float $x = \pm m 2^e$ becomes

$$2^q x = \pm m 2^{e+q}.$$

Only the exponent changes; no significand bit is rounded. The operation is exact whenever the result remains normal and representable, which is why the threshold $t = 2^{-2}$ is chosen conservatively to keep the trailing block inside the FP16 normal range.

Panel-diagonal proxy. Let T_k be the maximum trailing-block magnitude and d_k the maximum panel-diagonal magnitude. Figure 2 of the paper shows empirically that $T_k \leq C d_k$ with modest C . With

$$s = 2^{\lceil \log_2(d_k/t) \rceil},$$

we have $s \geq d_k/t$, so $d_k/s \leq t$ and

$$\frac{T_k}{s} \leq C \frac{d_k}{s} \leq C t.$$

This is the conditional statement Algorithm 3 relies on: *if* the trailing block tracks the panel diagonal, *then* the cheap $O(w)$ diagonal monitor controls the trailing magnitude as well.

What is and is not deterministic here. The exactness of 2^q -scaling is a deterministic floating-point fact. The bound $T_k \leq C d_k$ is empirical (Figure 2). The Section 3.5 statement is conditional on the empirical bound.

Summary

Rule	μ_k	α_k	Resulting ρ bound
PP	1	1	2^{n-1}
DP (success)	$1/\tau$	1	$(1 + 1/\tau)^s 2^{n-1-s}$
GP	$< \tau$	$\sqrt{c_k^{-2} - 1}$	$\prod_k (1 + \mu_k \sqrt{c_k^{-2} - 1})$
ScaP	1	1	2^{n-1}

Every formula in Section 3 is obtained by filling in the relevant row of this table and multiplying through.